Math 135 **Homework 2**
UCLA, Fall 2025 **due 10/15/2025 at 11:59pm**

Directions. Complete all the problems. Your solutions to the problems should be submitted to Gradescope before the indicated due date above. Please follow rules regarding Gradescope submission as described in the syllabus. If you use any outside source in your solutions, you must cite the source.

Problem 1. Compute each of the following

(a) $L\left[\dfrac{\sin x}{x}\right]$

(b) $L\left[t^n e^{at}\right]$ where a is a real number and n is a positive integer

Problem 2. Fully compute each of the following:

(a) $L^{-1}\left[\dfrac{1}{p^2(p^2+k^2)}\right]$

(b) $L^{-1}\left[\dfrac{1-2p}{p^2+4p+5}\right]$

(c) $L^{-1}\left[\dfrac{8p^2-4p+12}{p(p^2+4)}\right]$

Problem 3. Solve the following IVP's:

(a) $y''-2y'+2y=\cos t; \qquad y(0)=1, \quad y'(0)=0$

(b) $y''+2y'+y=4e^{-t}; \qquad y(0)=2, \quad y'(0)=-1$

Problem 4. Solve $\begin{cases} y^{(4)}-4y'''+6y''-4y'+y=0 \\ y(0)=0 \quad y'(0)=1 \quad y''(0)=0 \quad y'''(0)=1 \end{cases}$

Problem 5. Consider $L^{-1}\left[\dfrac{p}{(p^2+1)^2}\right]$

(a) Compute it using convolution.

(b) Compute it without using convolution.

Problem 6. Solve $y'' - 2y' + y = f(t), \quad y(0) = 0 \quad y'(0) = 1$

where $f(t) = \begin{cases} 0 & 0 \le t < 1 \\ t & 1 \le t < 2 \\ 0 & 2 \le t < \infty \end{cases}$

Problem 7. Solve $\begin{cases} y'' - 5y' + 6y = g(t) \\ y(0) = 0 \\ y'(0) = 0 \end{cases}$

Problem 8. Solve $\begin{cases} y'' + 9y = \delta(x - 3\pi) + \cos(3x) \\ y(0) = 0 \\ y'(0) = 0 \end{cases}$

Problem 9. Solve the Volterra integral equation

$$f(t) + 2\int_0^t \cos(t - x) f(x) dx = e^{-t}$$